THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · EASY LEVEL CHAPTER TEST SET 2

Chapter 10: **How Quantities Combine (Understanding Data)** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In a 100% stacked bar of a household's monthly electricity use, appliances take 32% of 100 units. The units used by the appliances are: 1

(a) 32 **(b)** 31 **(c)** 34 **(d)** 33

2. A stacked bar for a family's monthly food use shows rice 33 kg, wheat 32 kg, pulses 33 kg. The total length of the bar is (in kg): 1

(a) 99 **(b)** 97 **(c)** 100 **(d)** 98

3. The average marks of 24 students is 41 and of another 12 students is 44. The average of all 36 students is: 1

(a) 42 **(b)** 44 **(c)** 43 **(d)** 41

4. The concentration of salt in 400 mL of brine is 5%. The salt present is: 1

(a) 40 mL **(b)** 20 mL **(c)** 2 mL **(d)** 5 mL

5. The plain mean of two group averages equals the combined average only when: 1

(a) the averages are equal only **(b)** the groups have different sizes
(c) the two groups have the same size **(d)** one average is zero

6. Marks of 85, 51 and 79 are given weights 2, 3 and 1 respectively. The weighted average is: 1

(a) 67 **(b)** 68 **(c)** 69 **(d)** 66

7. The mean of 12, 11, 45, 28, 24 is: 1

(a) 23 **(b)** 25 **(c)** 24 **(d)** 26

8. Over 5 weeks of equal length a cyclist's weekly average distances were 43, 32, 50, 40, 35 km per day. The average over the whole period (km per day) is: 1

(a) 39 **(b)** 41 **(c)** 40 **(d)** 42

9. In a 100% stacked bar of a household's monthly electricity use, appliances take 25% of 140 units. The units used by the appliances are: 1

(a) 34 **(b)** 37 **(c)** 36 **(d)** 35

10. A stacked bar for a family's monthly food use shows rice 36 kg, wheat 16 kg, pulses 32 kg. The total length of the bar is (in kg): 1

(a) 86 **(b)** 85 **(c)** 84 **(d)** 83

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A 100% stacked bar shows that 30% of a family's monthly spending of ₹140 is on food. Find the amount spent on food and on everything else. 2

12. A 350 mL bottle of a drink contains 28% fruit pulp. Find the volume of pulp and of the remaining liquid. 2

13. Find the mean of the numbers 12, 17, 62, 42, 68, 15. 2

14. A 100% stacked bar shows that 15% of a family's monthly spending of ₹160 is on food. Find the amount spent on food and on everything else. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The average marks of 26 students in a test is 39 and that of 18 students is 61. Find the average marks of all 44 students together. 3

16. A girl cycles for 4 weeks of equal length. Her weekly averages are 53 min, 47 min, 59 min, 49 min per day. Find the average over all the weeks and say why the simple mean of the averages can be used. 3

17. A final score combines Maths (score 51, weight 2), Science (score 95, weight 2). Find the weighted average score. 3

OR

The average marks of 26 students in a test is 61 and that of 13 students is 58. Find the average marks of all 39 students together. 3

18. A girl cycles for 3 weeks of equal length. Her weekly averages are 50 min, 60 min, 34 min per day. Find the average over all the weeks and say why the simple mean of the averages can be used. 3

OR

A final score combines Maths (score 95, weight 2), Science (score 89, weight 2). Find the weighted average score. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The mean of 7 observations is 41. A new observation 57 is added. Find the new mean. 5

OR

40 mL of a 18% sugar solution is mixed with 20 mL of a 24% sugar solution. Find the percentage concentration of the mixture. 5

20. The mean of 9 observations is 44. A new observation 54 is added. Find the new mean. 5

OR

40 mL of a 27% sugar solution is mixed with 60 mL of a 22% sugar solution. Find the percentage concentration of the mixture. 5

END OF QUESTION PAPER — Total: 40 Marks